

Ai Workshop with Couchbase

Agenda

1. 名詞解釋
2. 環境安裝/設定
3. exercise 1 - 基礎操作
4. exercise 2 - AI客服
5. AI db 選擇, 為什麼選用Couchbase
6. Couchbase其他功能

AI 名詞解釋

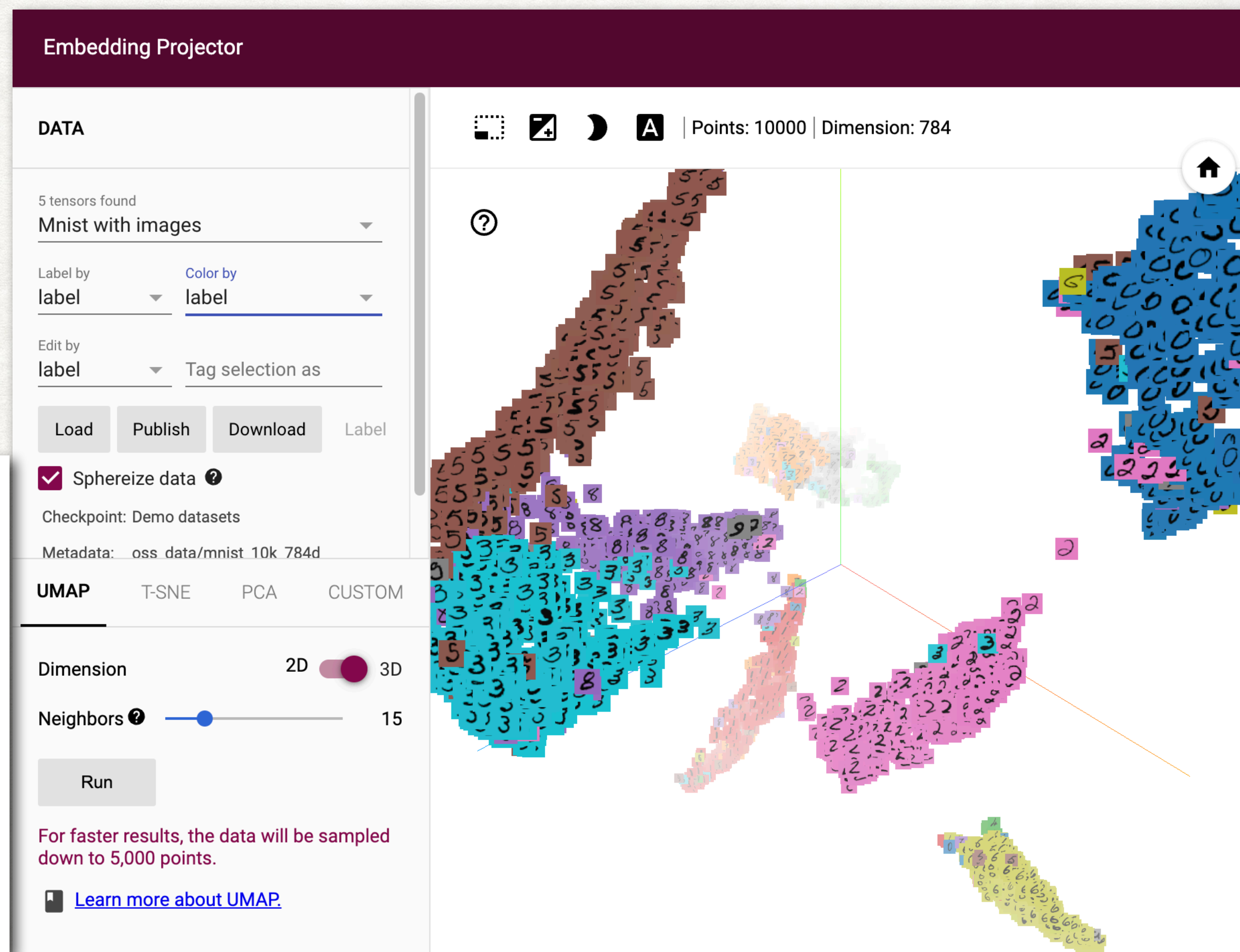
Embedding

- Embedding 是一種將原始數據轉換成一種特別的數據格式。
- 將語言、物體或其他類型的資料表示為固定維度的向量（數值組）。

"Ollama is the easiest way to get up and running with large language models."



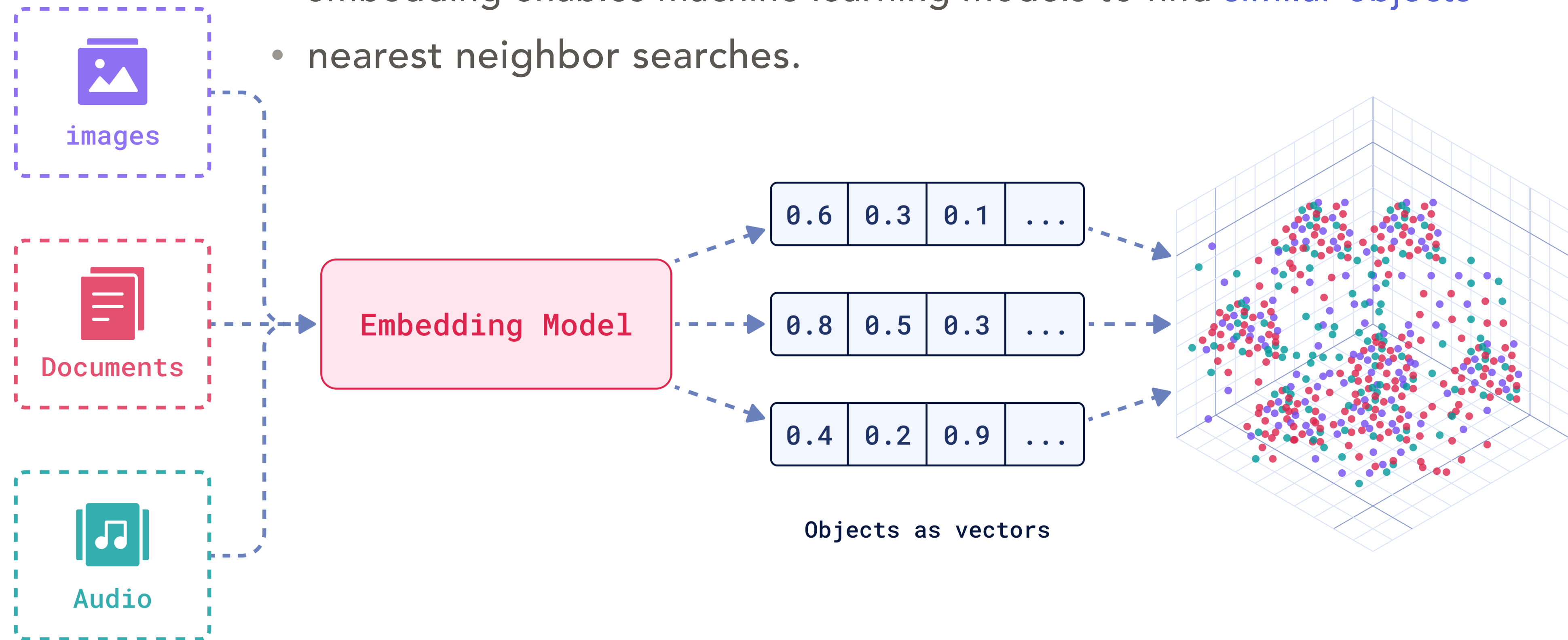
```
[ -0.15521588921546936,  
  -0.3130679428577423,  
  -0.2622824013233185,  
  -0.10730823874473572,  
  ...  
  0.26006409525871277,  
  0.14494779706001282,  
  -0.01514953002333641,  
  0.04403747618198395]
```



AI 名詞解釋

Why Embedding

- embedding enables machine learning models to find **similar objects**
- nearest neighbor searches.



AI 名詞解釋

Embedding Model



ExplorePricingDocs

 **daanelson / imagebind** 

A model for text, audio, and image embeddings in one space

• Warm

 Public

 3.3M runs

 [GitHub](#)

 [License](#)

Playground

API

Examples

README

Versions

Input

Form

JSON

Node.js

Python

HTTP

Cog

Docker



Add a payment method to run this model.

Each run costs approximately \$0.00022. Alternatively, try out our [featured models](#) for free.

 Sign in with GitHub

By signing in, you agree to our [terms of service](#) and [privacy policy](#)

Model	Costs encoding 1M docs	Performance
OpenAI GPT-3 Embedding Models		
text-similarity-ada	\$800	61.86
text-similarity-babbage	\$1,200	62.62
text-similarity-curie	\$6,000	62.39
text-similarity-davinci	\$60,000	58.11
Google Embedding Models		
st5-base-1	\$0.70	67.84
st5-large-1	\$2.40	68.74
st5-3b-1	\$6.80	69.23
universal-sentence-encoder-large-5	\$0.35	64.51
Sentence-Transformers Model		
all-MiniLM-L6-v2	\$0.12	68.06
all-MiniLM-L12-v1	\$0.25	68.83
all-mpnet-base-v1	\$0.70	69.98
all-roberta-large-v1	\$2.40	70.23

Output

PreviewJSON

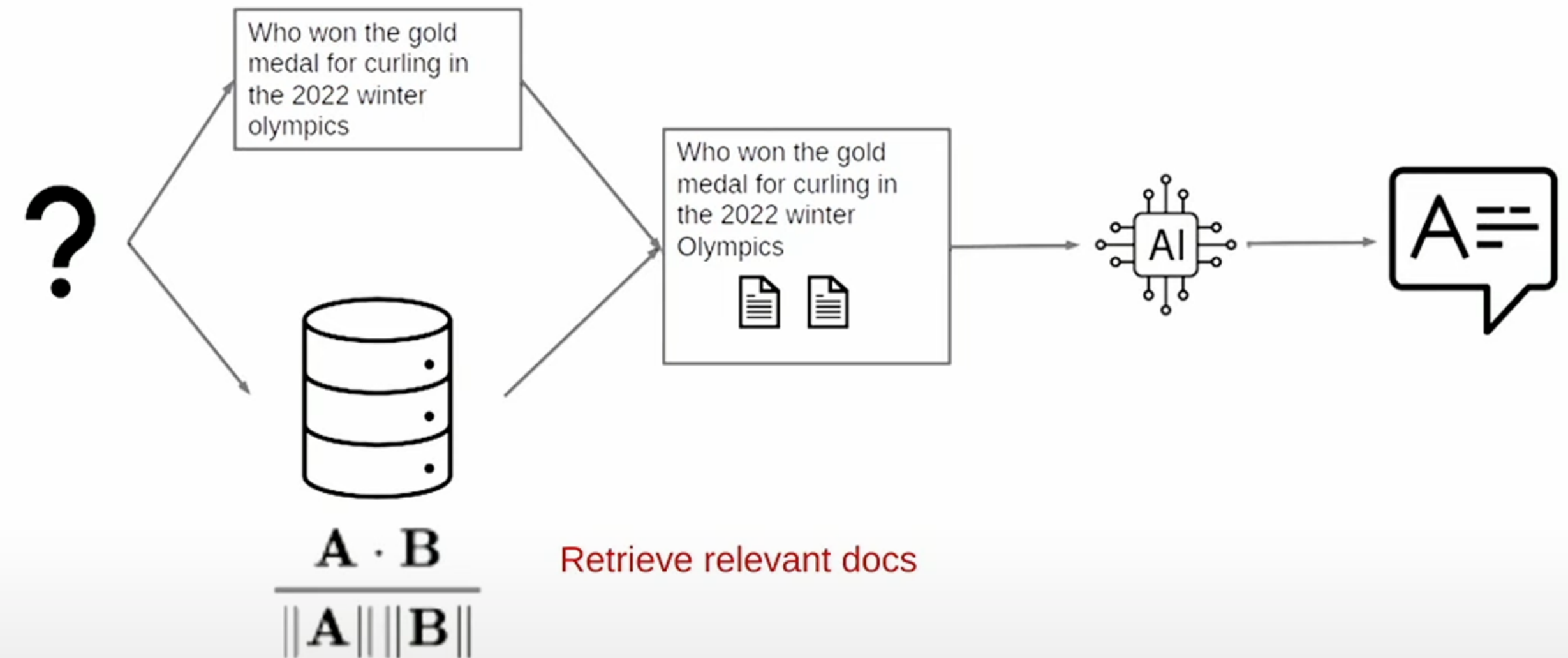
```
[  
  -0.04028015583753586,  
  0.032599665224552155,  
  0.0296342670917511,  
  0.017571788281202316,  
  0.00795517023652792,  
  0.04293808341026306,  
  0.014025846496224403,  
  -0.02359229139983654,  
  0.033283885568380356,  
  -0.027835514396429062,  
  0.006876518949866295,  
  0.0124134486541152,  
  -0.011771892197430134,  
  -0.023389752954244614,  
  -0.01167969685047865,  
  -0.01347434800118208,  
  -0.010429474525153637,  
  -0.005077551305204037
```


AI 名詞解釋

RAG

- 傳統的生成模型只能依賴訓練期間學到的知識。
- 生成模型有時會捏造不存在的答案

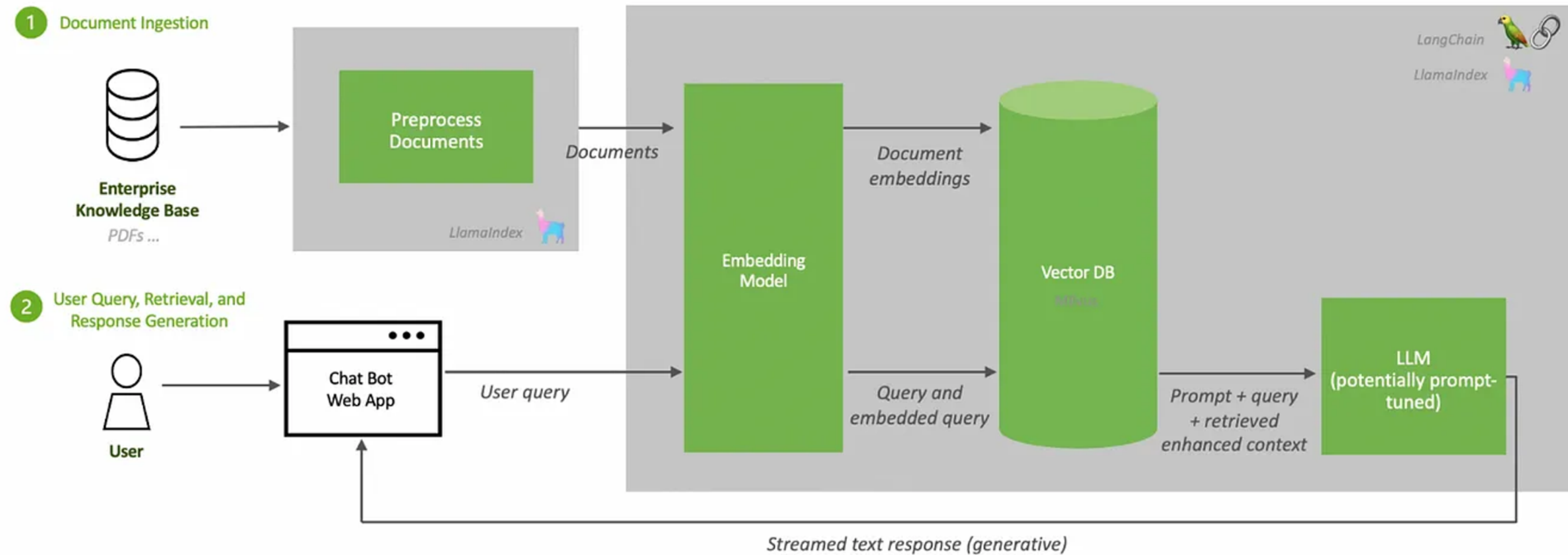
Retrieval Augment Generation - Q & A



AI 名詞解釋

RAG

Retrieval Augmented Generation (RAG) Sequence Diagram



AI 名詞解釋

Prompt

- 一個問題，一個敘述，或者甚至是一個故事的開始

```
# Prompt
prompt_text = """You are an assistant tasked with summarizing tables and text. \
Give a concise summary of the table or text. Table or text chunk: {element} """
prompt = ChatPromptTemplate.from_template(prompt_text)
```


環境安裝

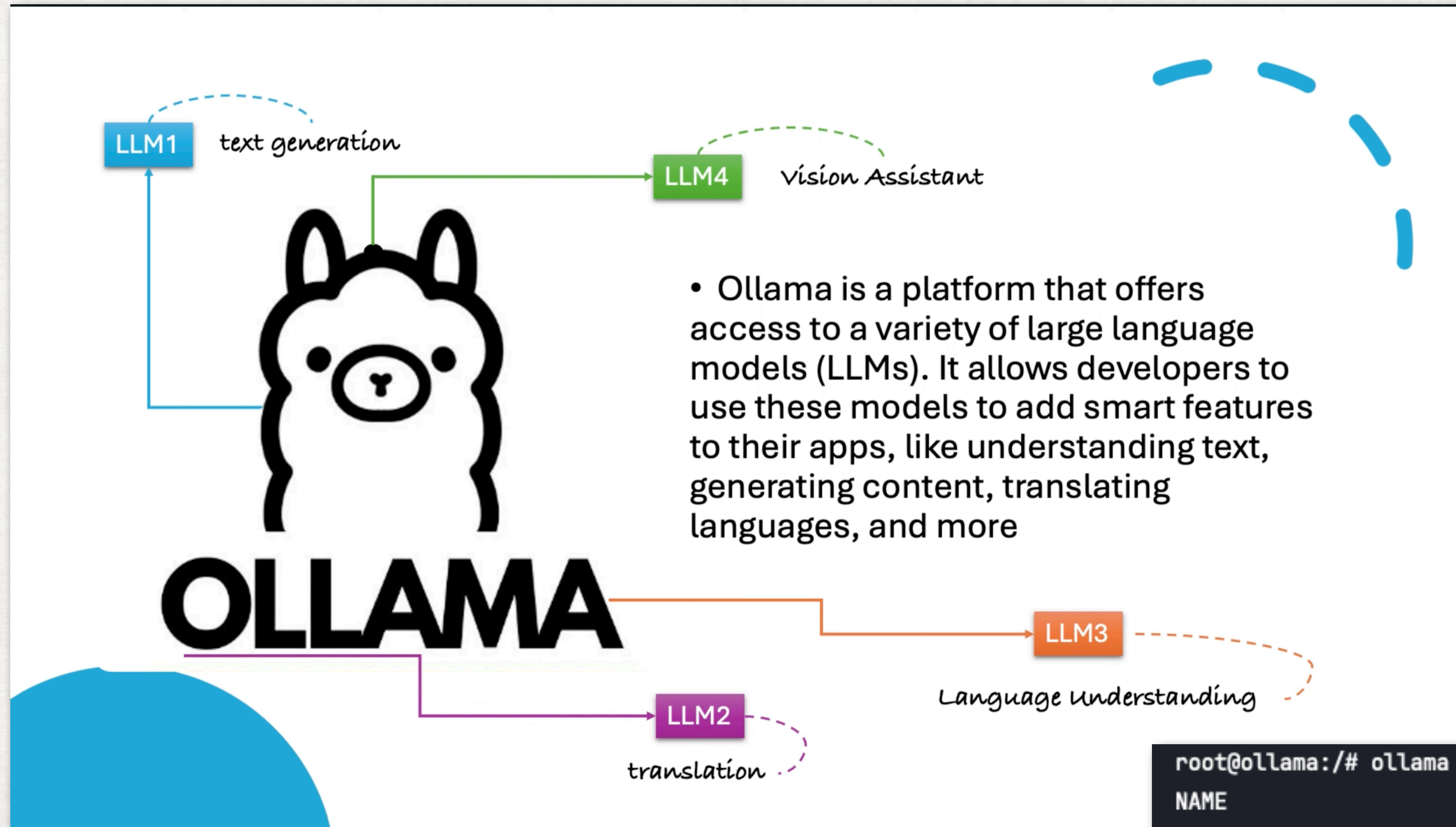
Docker compose

- `docker run -d --name couchbase -p 8091-8096:8091-8096 -p 11210-11211:11210-11211 couchbase:enterprise-7.6.1`

```
🚀 docker-compose.yml
1  services:
2    ollama:
3      hostname: ollama
4      container_name: ollama
5      image: ollama/ollama
6      ports:
7        - "11434:11434"
8      volumes:
9        - ollama-local:/root/.ollama
10
11   openWebUI:
12     container_name: open-webui
13     image: ghcr.io/open-webui/open-webui:main
14     restart: always
15     depends_on:
16       - ollama
17     ports:
18       - 3000:8080
19     extra_hosts:
20       - "host.docker.internal:host-gateway"
21     volumes:
22       - open-webui:/app/backend/data
23
24   volumes:
25     ollama-local:
26     open-webui:
```


Ollama

- 運行和管理本地LLM的平台。



<https://ollama.com/library>

<https://github.com/ollama/ollama>

```
root@ollama:/# ollama list
NAME                                ID                                SIZE    MODIFIED
moondream:latest                    55fc3abd3867                      1.7 GB  23 hours ago
llava-phi3:latest                    c7edd7b87593                      2.9 GB  2 days ago
nomic-embed-text:latest              0a109f422b47                      274 MB  3 days ago
qwen2:1.5b                           f6daf2b25194                      934 MB  4 days ago
qwen2:0.5b                           6f48b936a09f                      352 MB  4 days ago
```


openWebUI

←

→

↺

🏠

localhost:3000

📄

🔍

☆

☰

W

完成更新

OI 新增對話

✎

☰

🔧 工作區

🔍 搜尋

過去 7 天

你好

Hello

Workshop for Creativity and Under

💡 Workshop Details

🎨 Artistic Creative Workshop

Digital Workshop

🌟 Hello Universe

👋 Chinese Greetings 🇨🇳

🎉 Great Introductions! Its about to

Hey! How can I assist you today?

W willy

qwen2:0.5b

▼

+

設為預設

☰

W

OI

您好，Willy

今天我能為您做些什麼？

⚡ 建議

Help me study

vocabulary for a college entrance exam

提示詞

↑

Grammar check

rewrite it for better readability

提示詞

↑

Give me ideas

for what to do with my kids' art

提示詞

↑

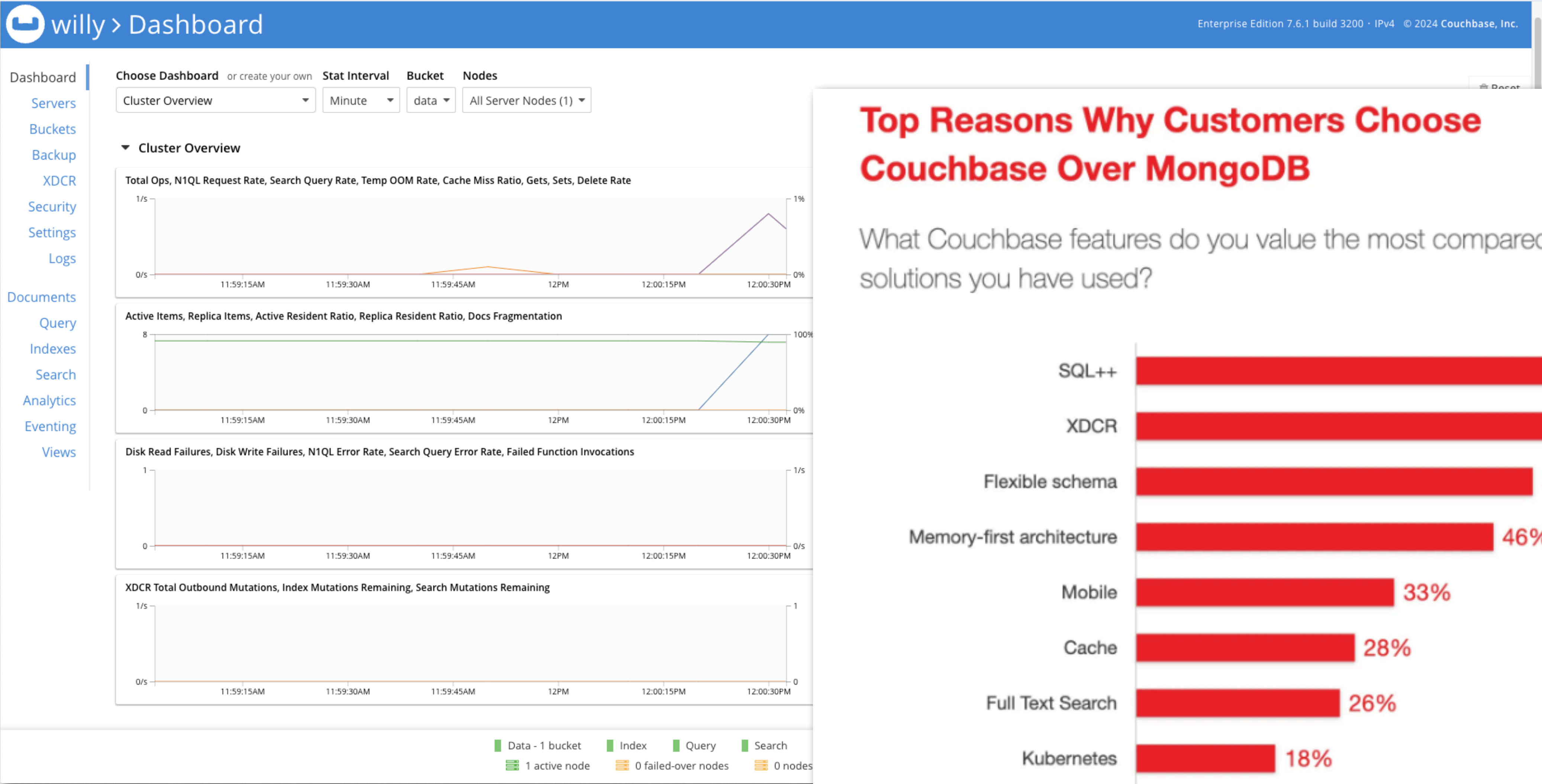
+ 傳送訊息

🎤

🎧

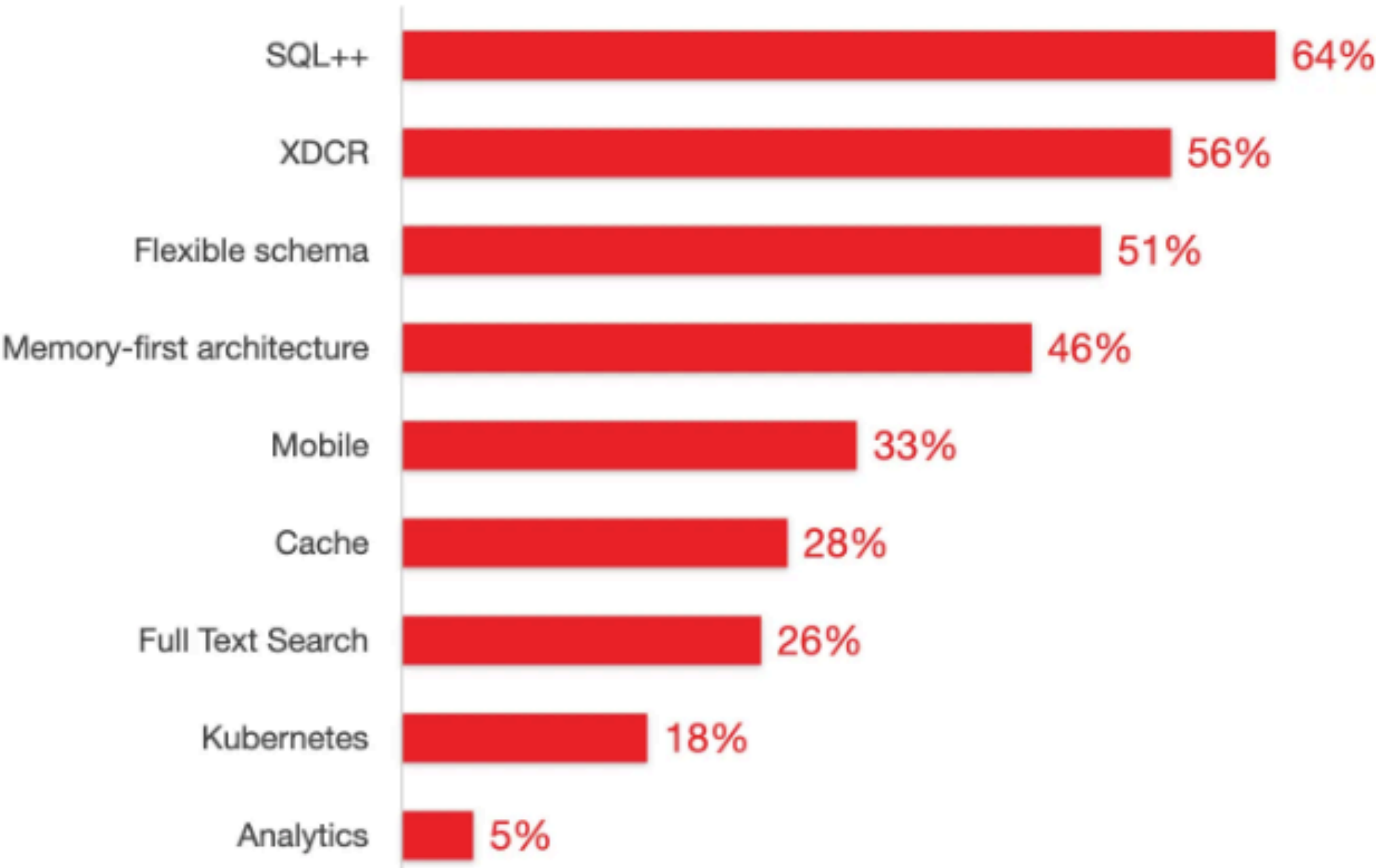
大型語言模型可能會出錯。請驗證重要資訊。

Couchbase



Top Reasons Why Customers Choose Couchbase Over MongoDB

What Couchbase features do you value the most compared to other solutions you have used?



exercise 1

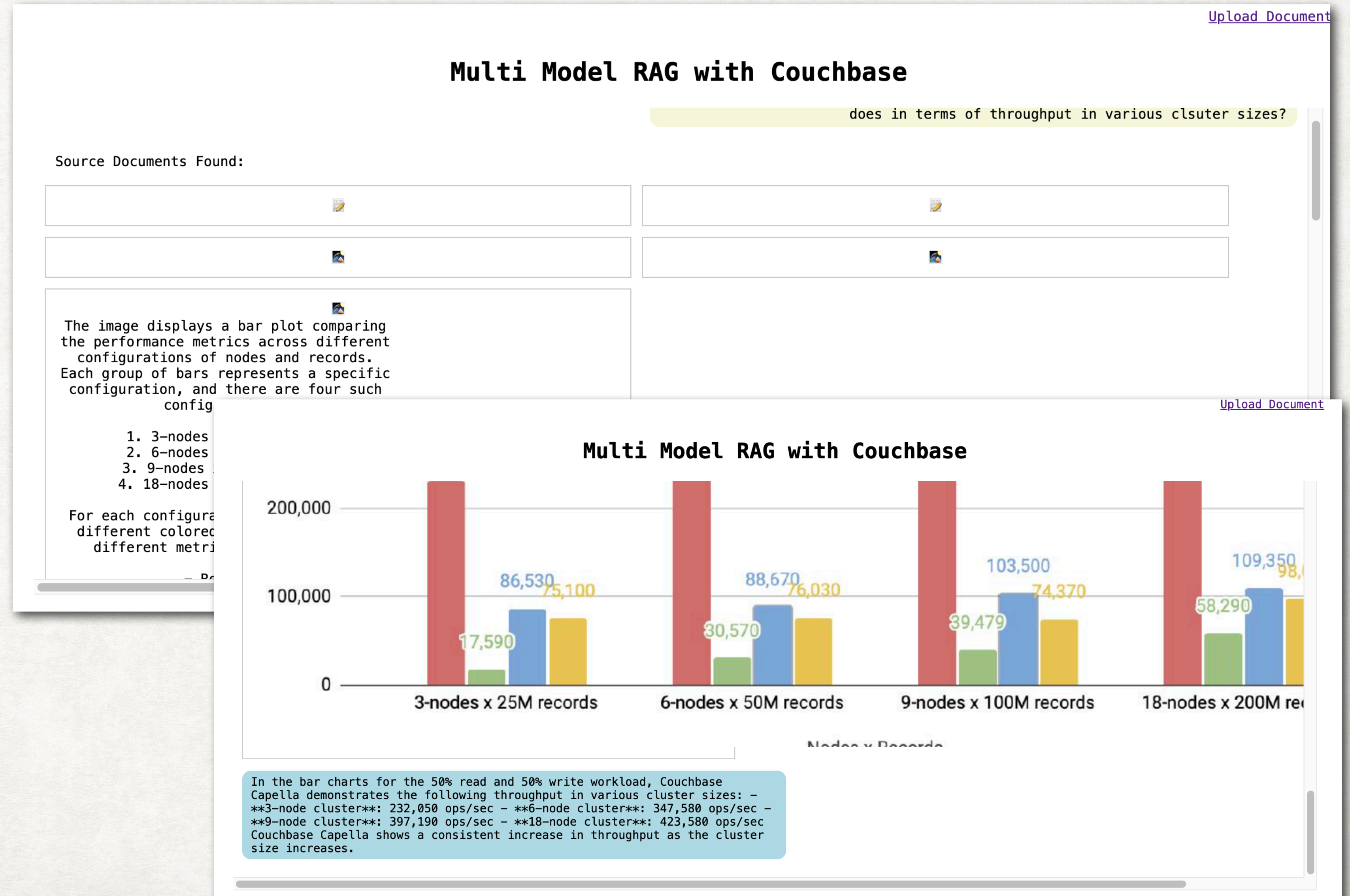
Flow

1. 透過程式與ai對話
2. 建立embedding
3. store data with embedding
 - insert json doc into couchbase
 - Select doc from couchbase
4. vector search
 - full-text search
5. get result by provied content

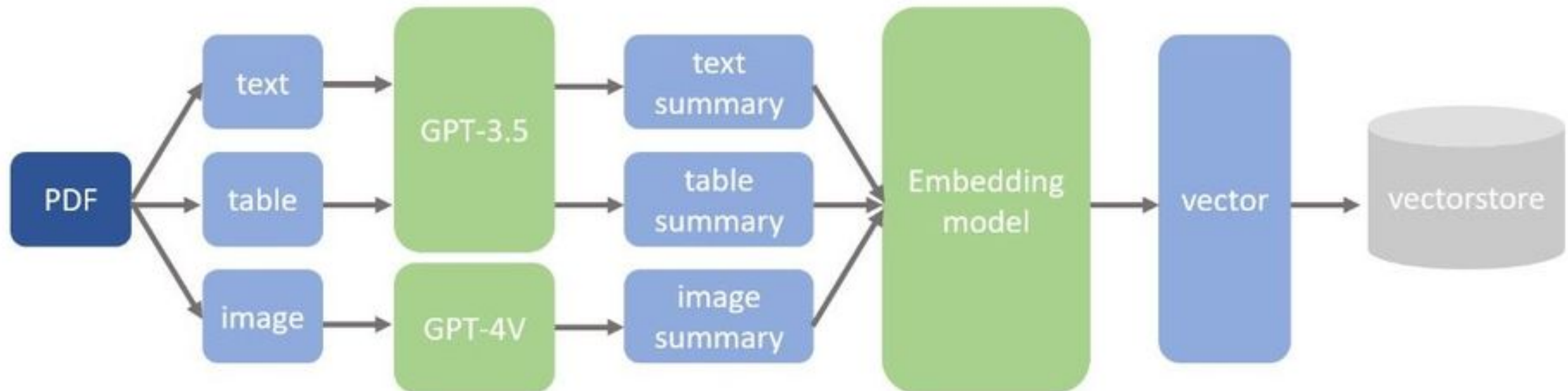
exercise 2

目標

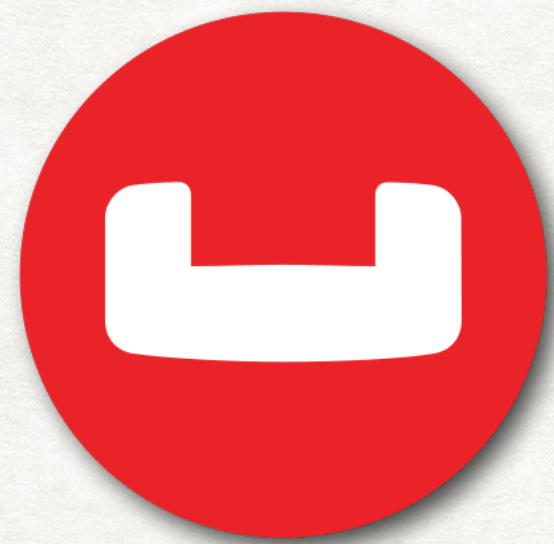
- 上傳pdf後, 請ai回答
相關問題



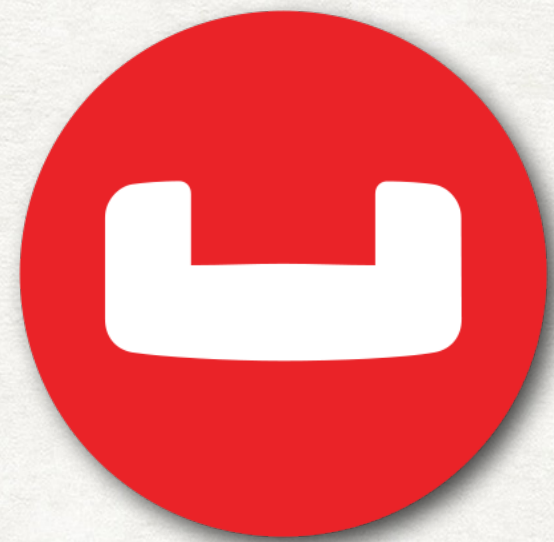
PDF partition



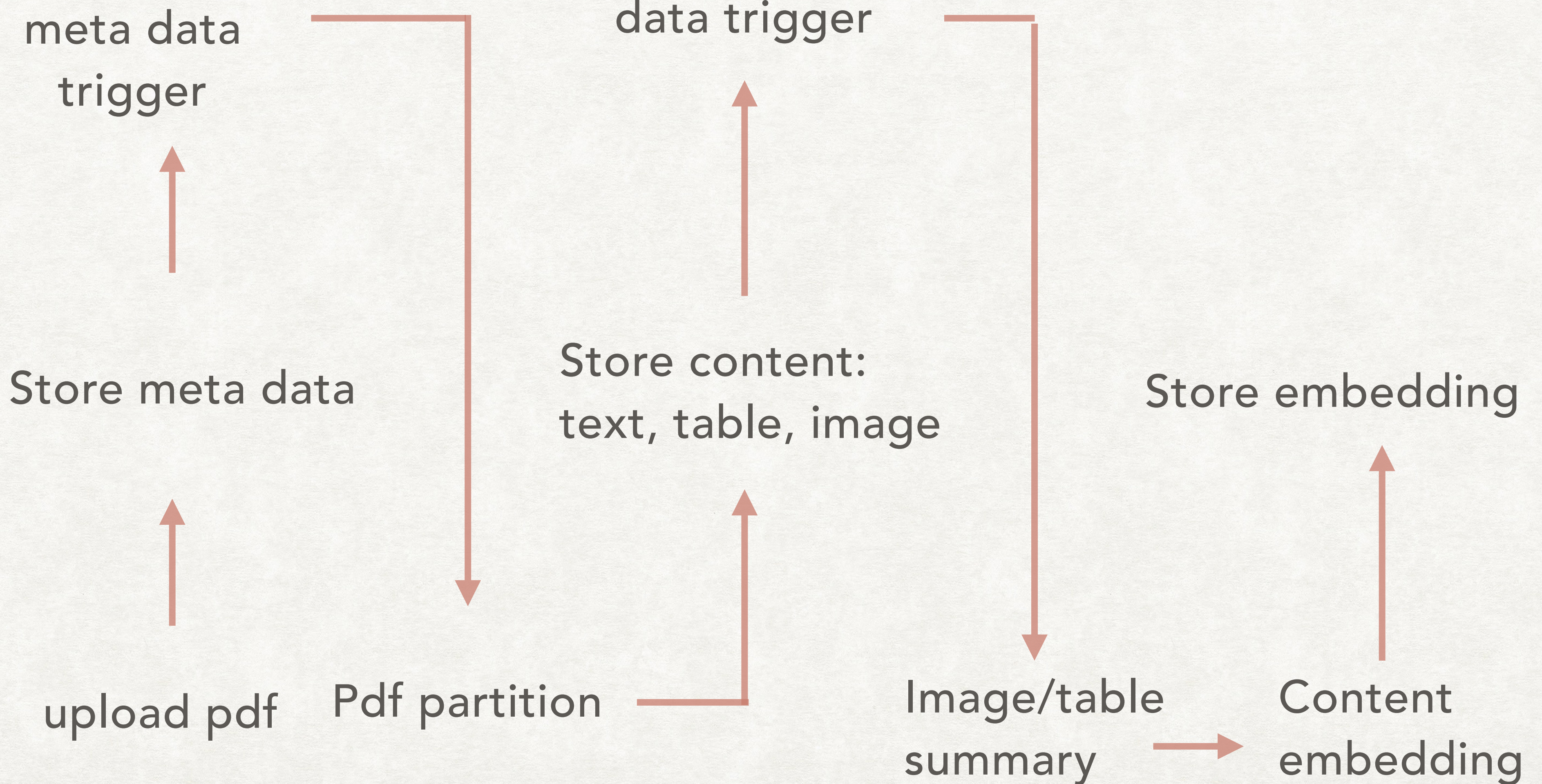
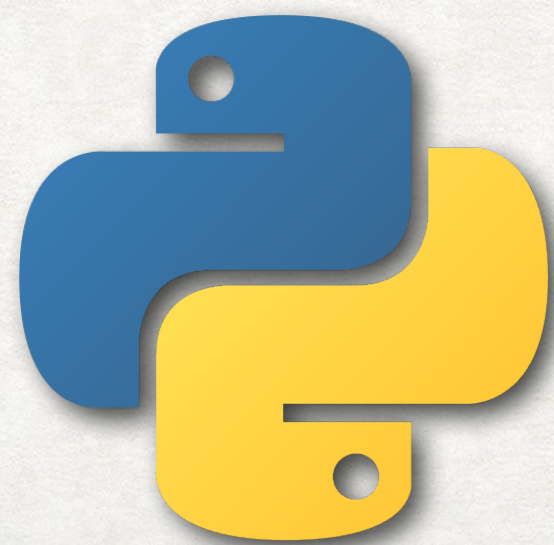
Prepare data



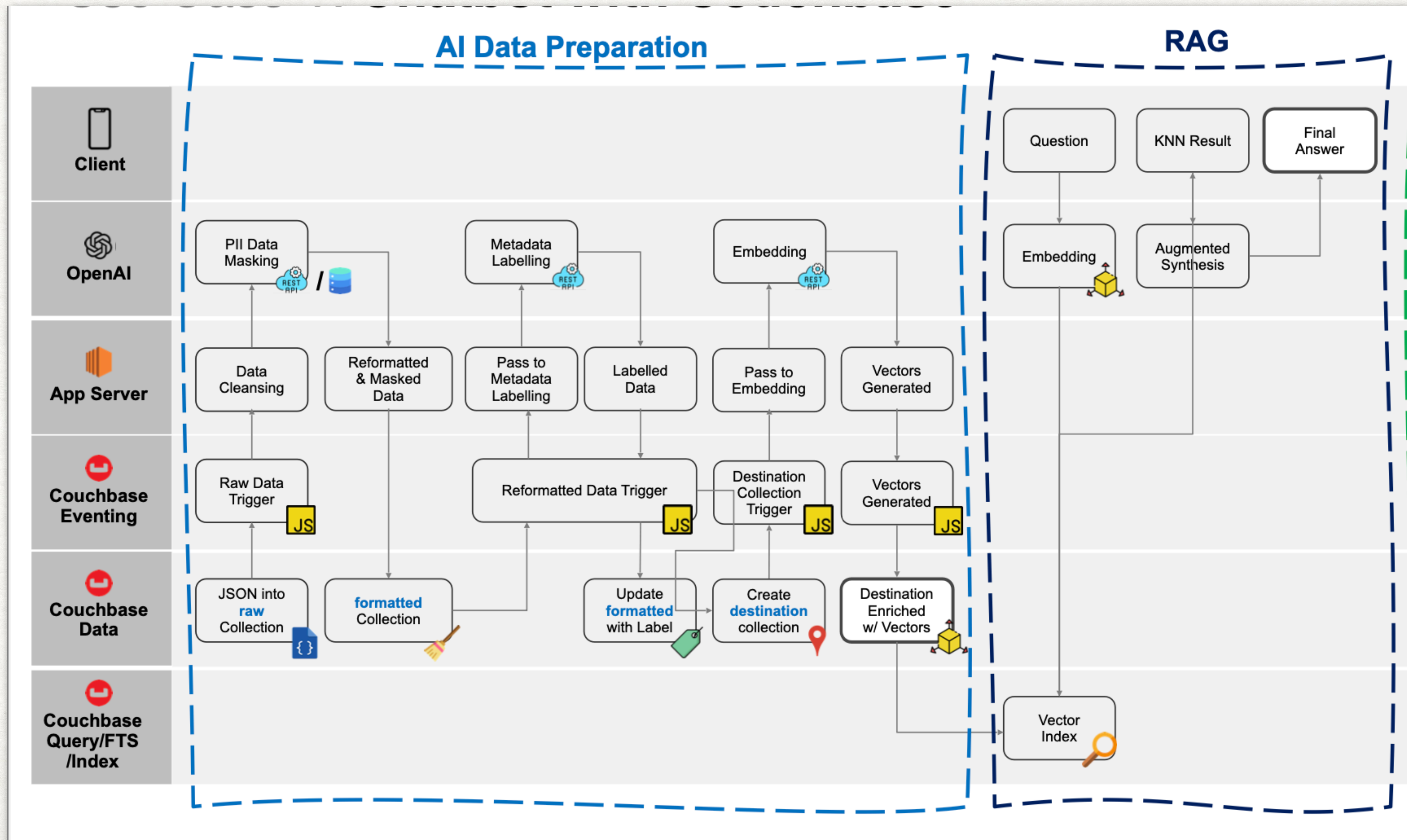
Eventing



Data



RAG



AI DB 選擇

- vector db
- 讀取寫入效能
- 擴展性
- 穩定性
- 備份和災害復原
- 成本：維運, 學習, 開發, 每單位資料
- 整合能力: raw data預處理, 與其他系統的整合, e.g. Kafka

AI DB 選擇

vector db

ecom > Full Text Search > Edit Index

← BACK

Dashboard

Servers

Buckets

Backup

XDCR

Security

Settings

Logs

Documents

Query

Indexes

Search

Analytics

Eventing

Views

Index Name

search-data

Bucket

data

Data ingest from:

• Scope: data

• Collections: ["data"]

Customize Index

☒ Use non-default scope/collection(s)

Scope

data

Type Identifier

Mappings

☒ # data.data | dynamic

category | text | index | store | include in_all field | include term vectors | docvalues

content | text | index | store | include in_all field | include term vectors | docvalues

field

embeddings

type

vector

searchable as

embeddings

dimension

1536

similarity metric

l2_norm

optimized for

recall

ok

cancel

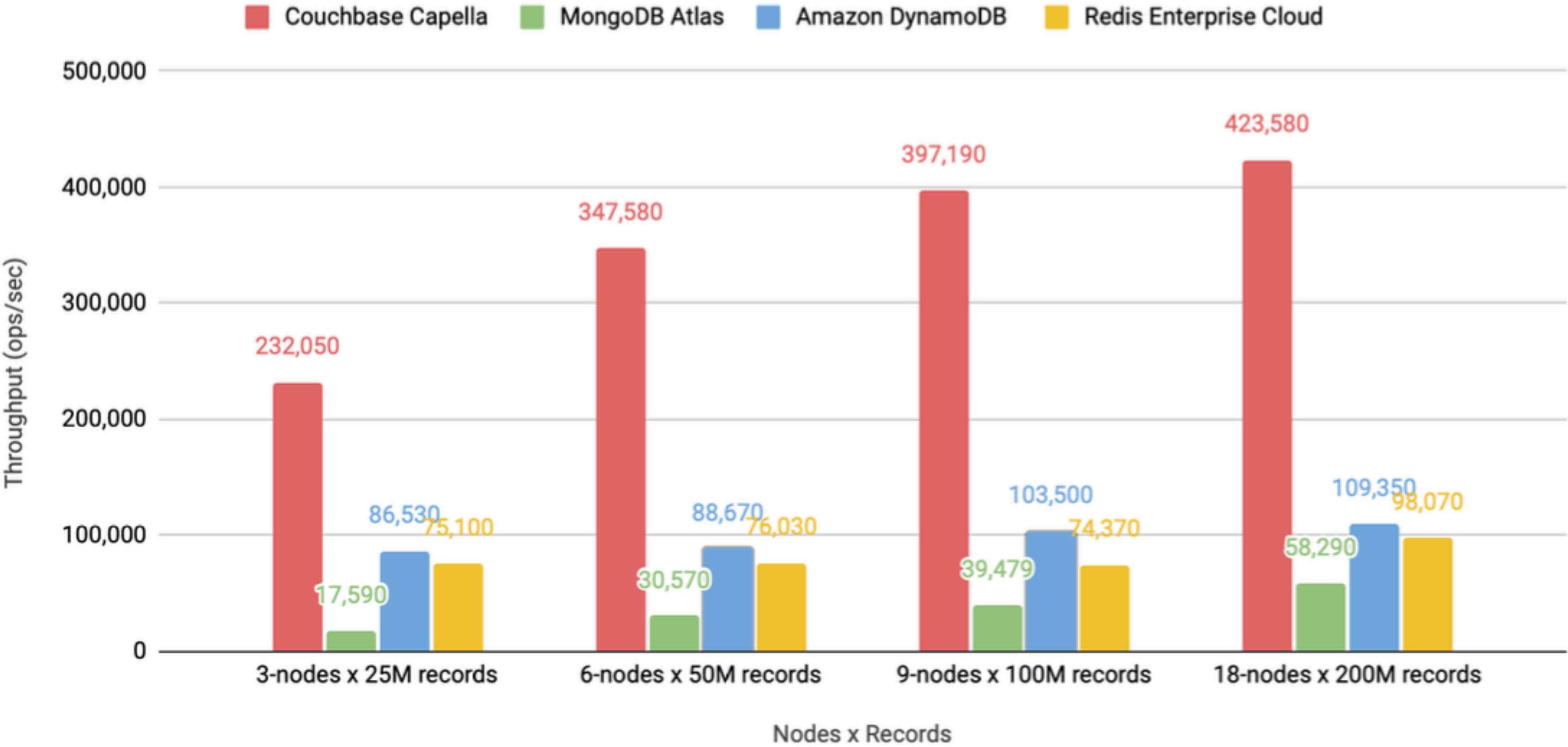
delete

Index Definition Preview

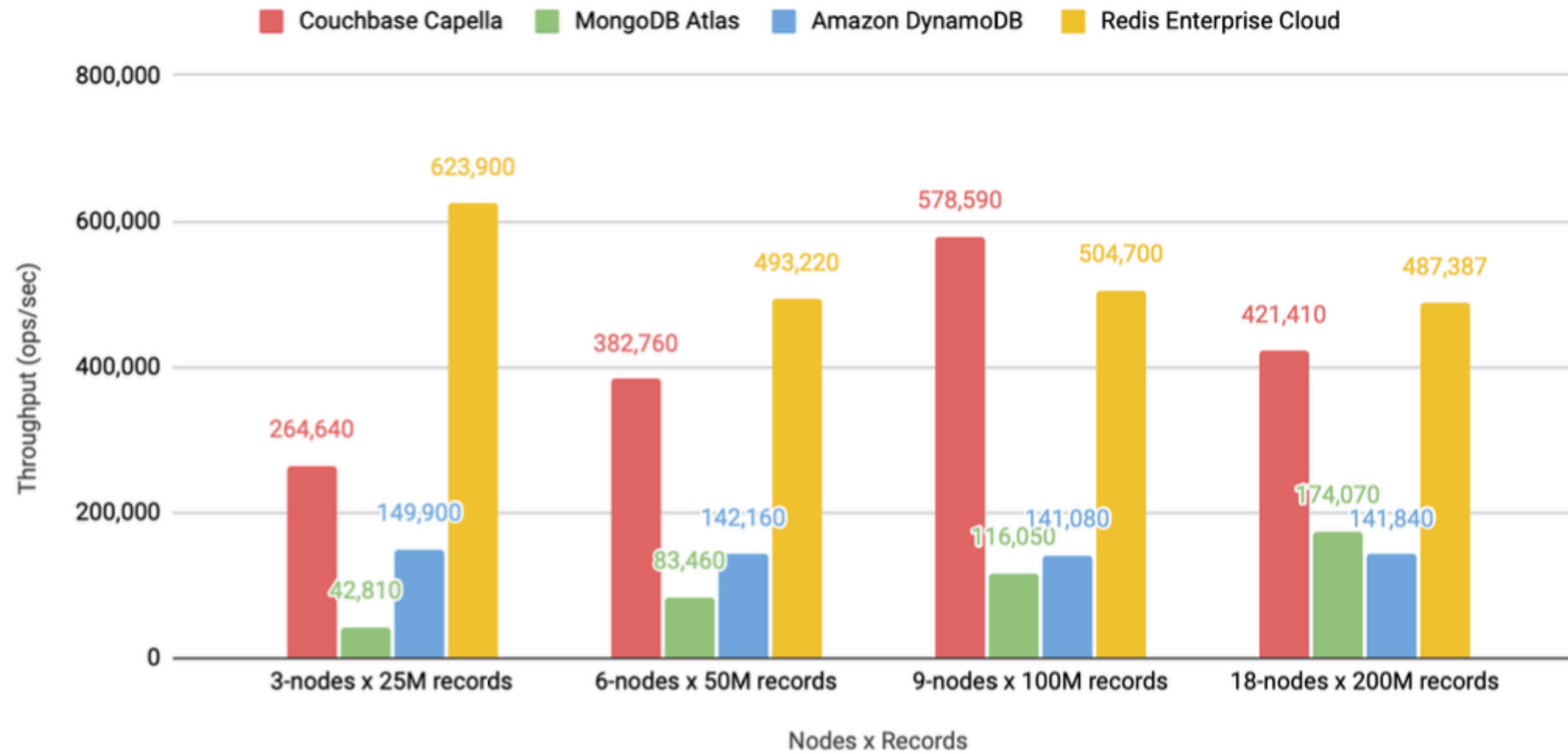
copy to clipboard

```
{
  "name": "search-data",
  "type": "fulltext-index",
  "params": {
    "doc_config": {
      "docid_prefix_delim": "",
      "docid_regexp": "",
      "mode": "scope.collection.type_f",
      "type_field": "type"
    },
    "mapping": {
      "default_analyzer": "standard",
      "default_datetime_parser": "date",
      "default_field": "_all",
      "default_mapping": {
        "dynamic": true,
        "enabled": false
      },
      "default_type": "_default",
      "docvalues_dynamic": false,
      "index_dynamic": true,
      "store_dynamic": false,
      "type_field": "_type",
      "types": {
        "data.data": {
          "dynamic": true,
          "enabled": true,
          "properties": {
```

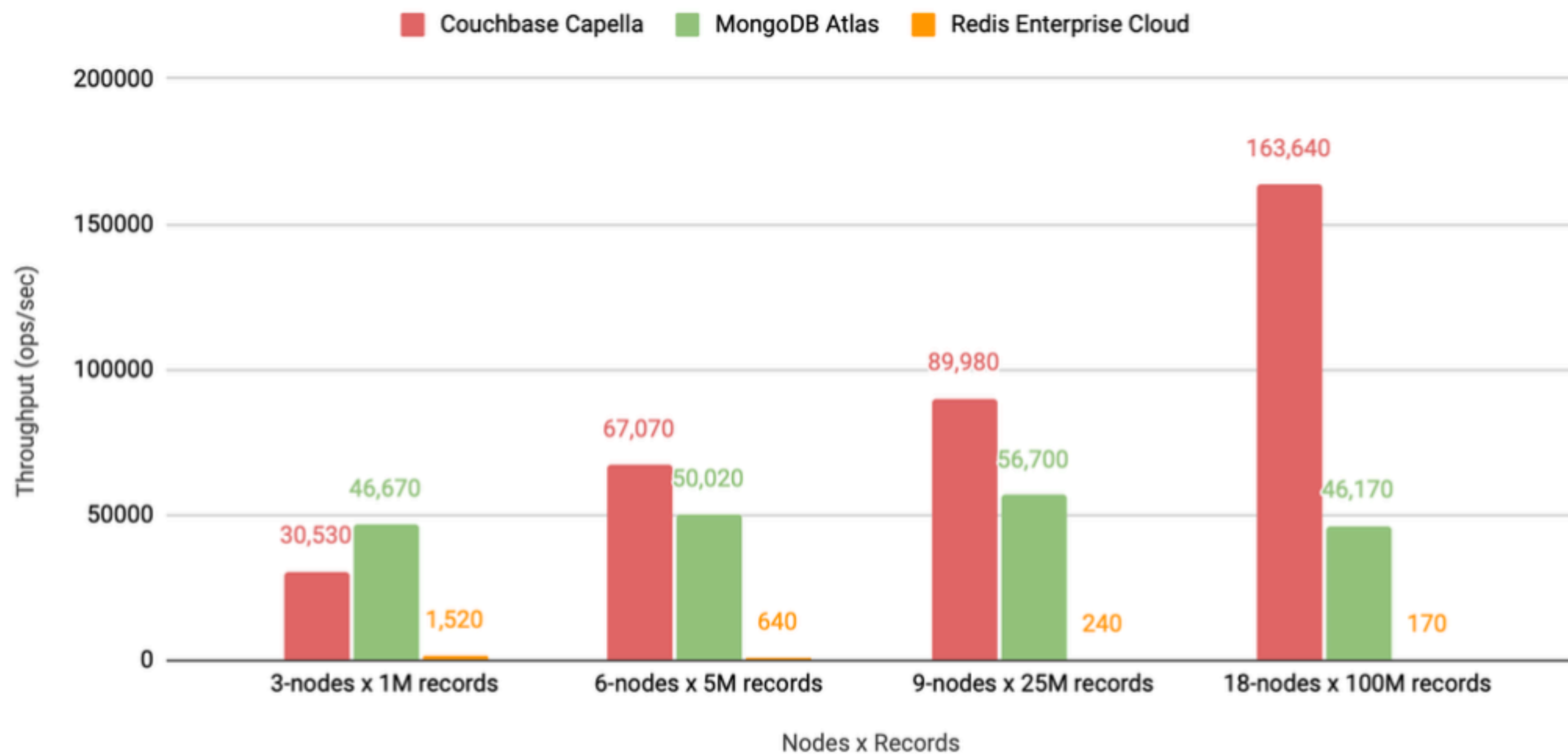

Workload A: 50% read & 50% update



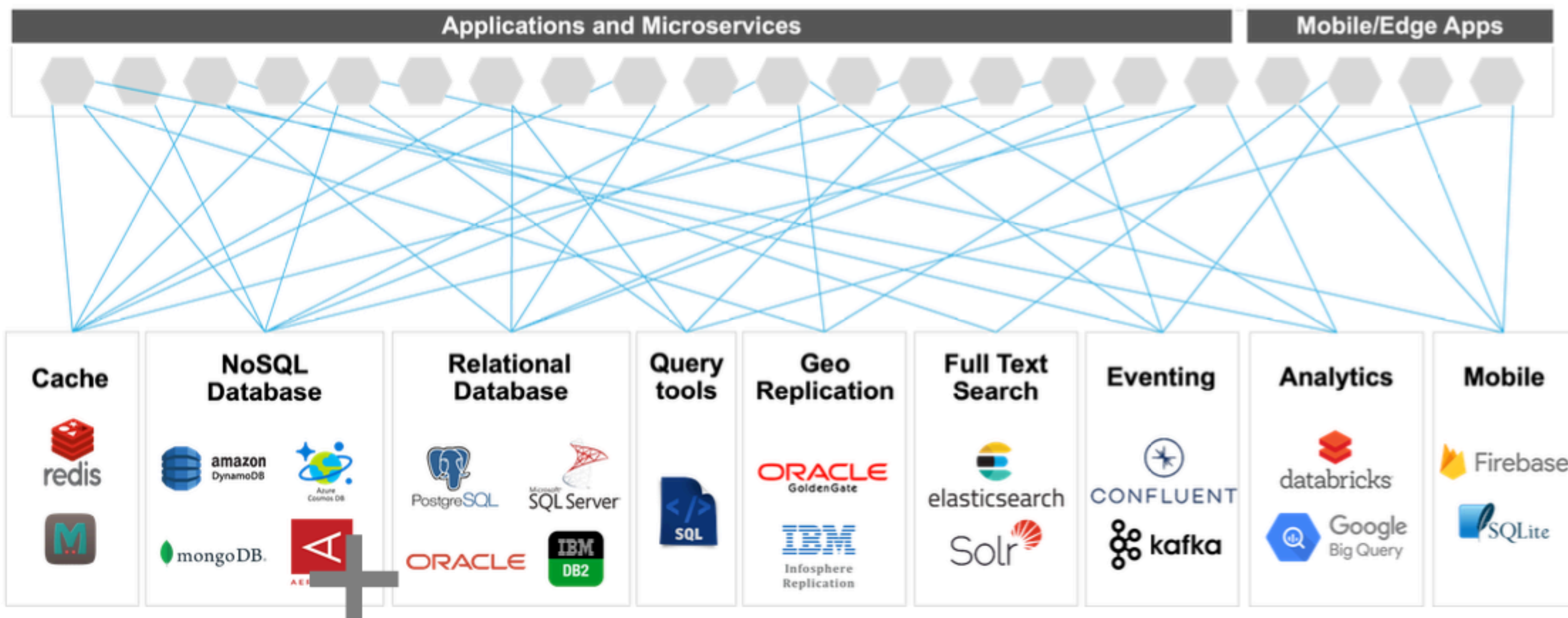
Workload C: 100% read



Pagination Workload: 100% read

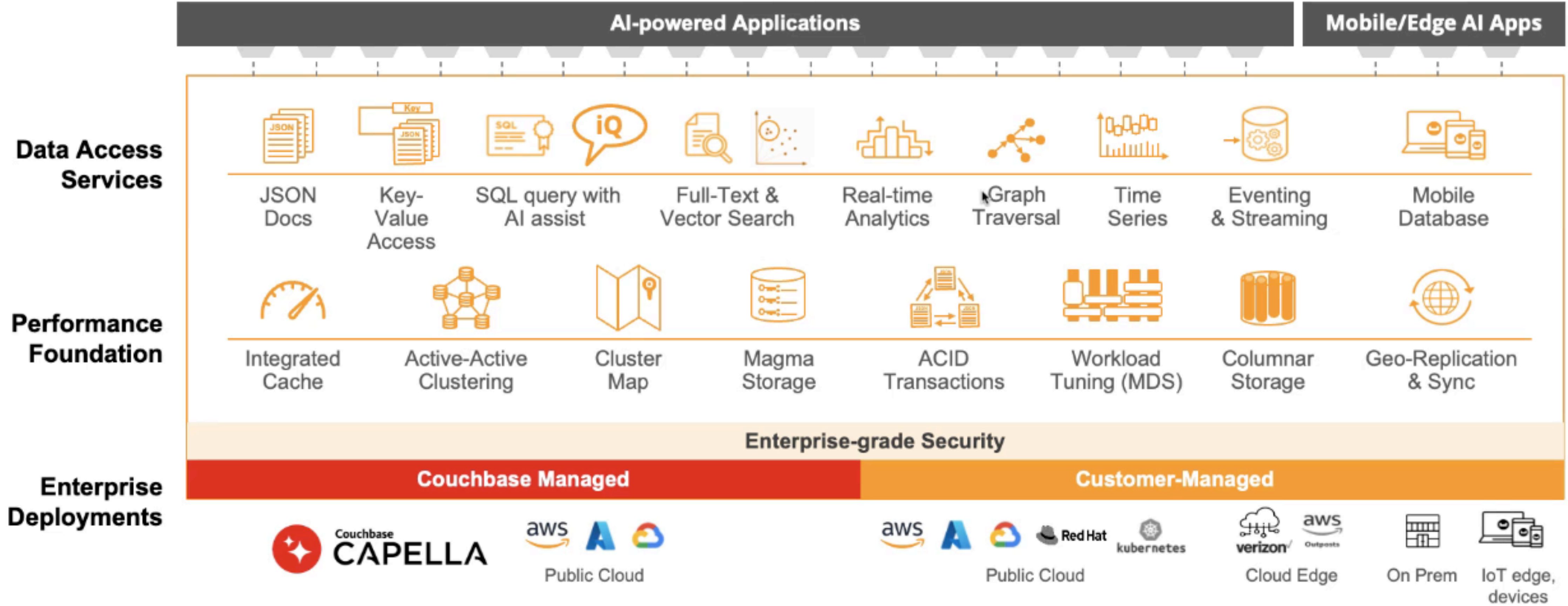


Couchbase 功能

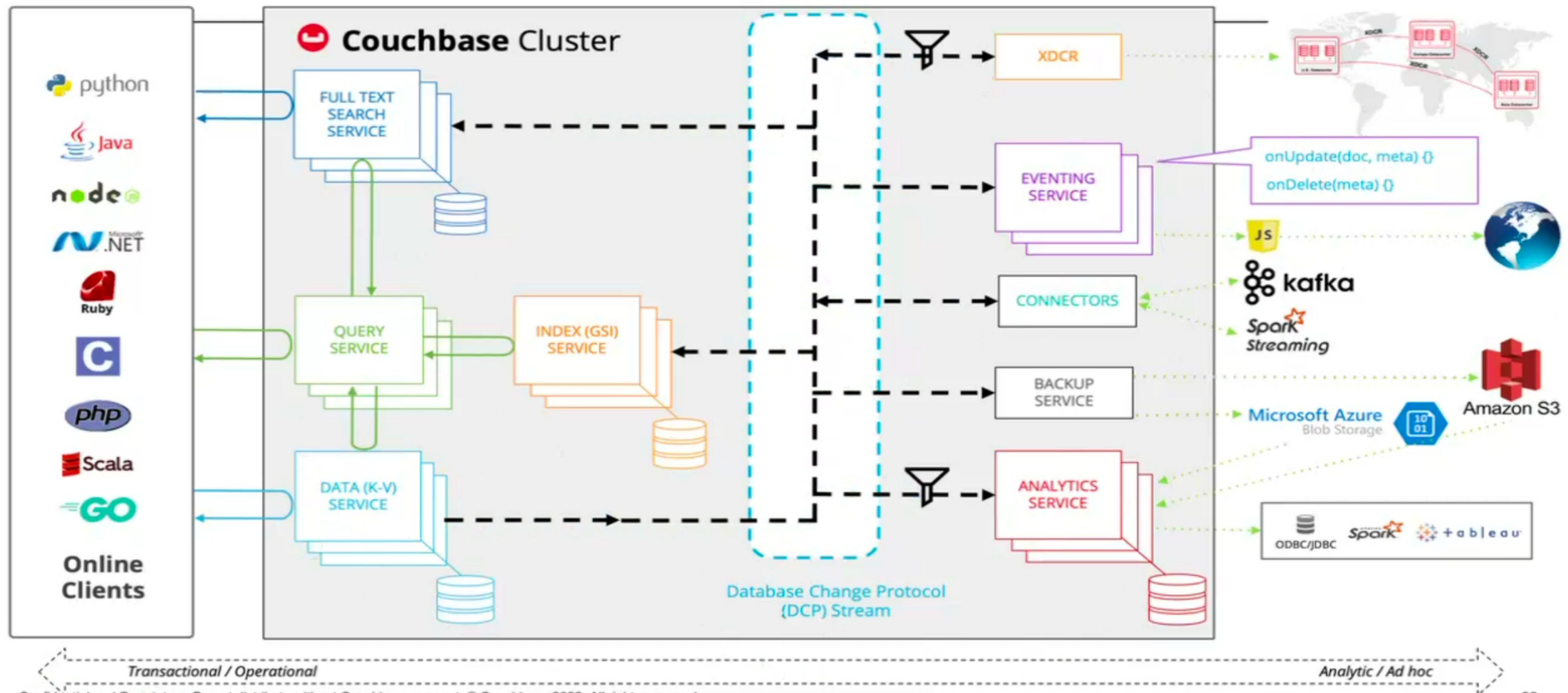


Couchbase's Differentiated Architecture

Fast, Affordable, Versatile, and Easy as SQL



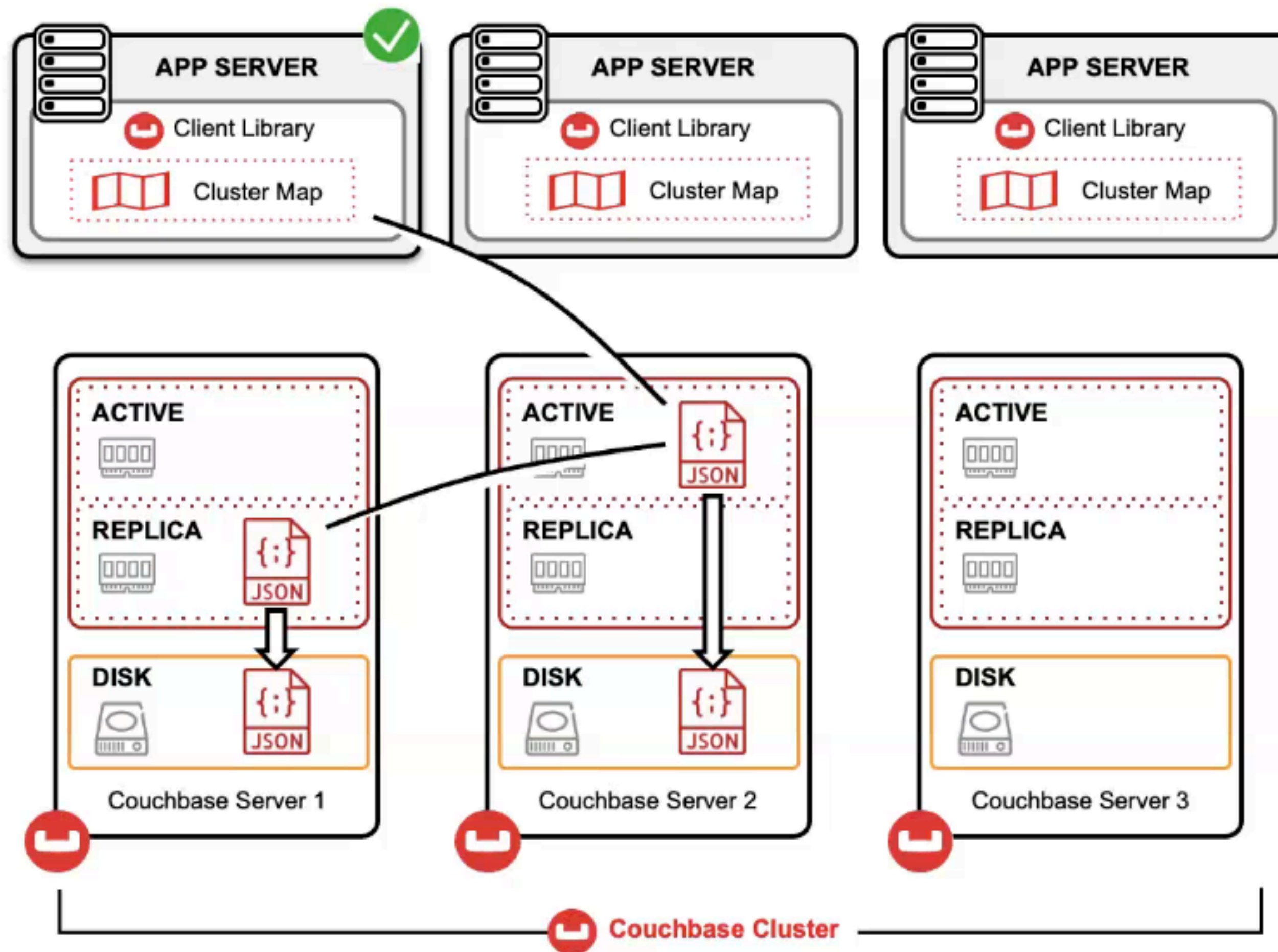
Services Architecture



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22

Durability



replica = 1

Level of Durability

None

Majority

MajorityAndPersistToActive

PersistToMajority

CENTRALIZED CLOUD



Capella / Couchbase Server App Services / Sync Gateway

Data processing
+ vector search

Sync Gateway ↔ Sync Gateway ↔ App Services

EDGE DATA CENTER



Couchbase Server

Couchbase Sync Gateway

Data processing
+ vector search

Couchbase Lite ↔ Sync Gateway

EDGE DEVICES



Smartphone



Tablet



Laptop



IoT

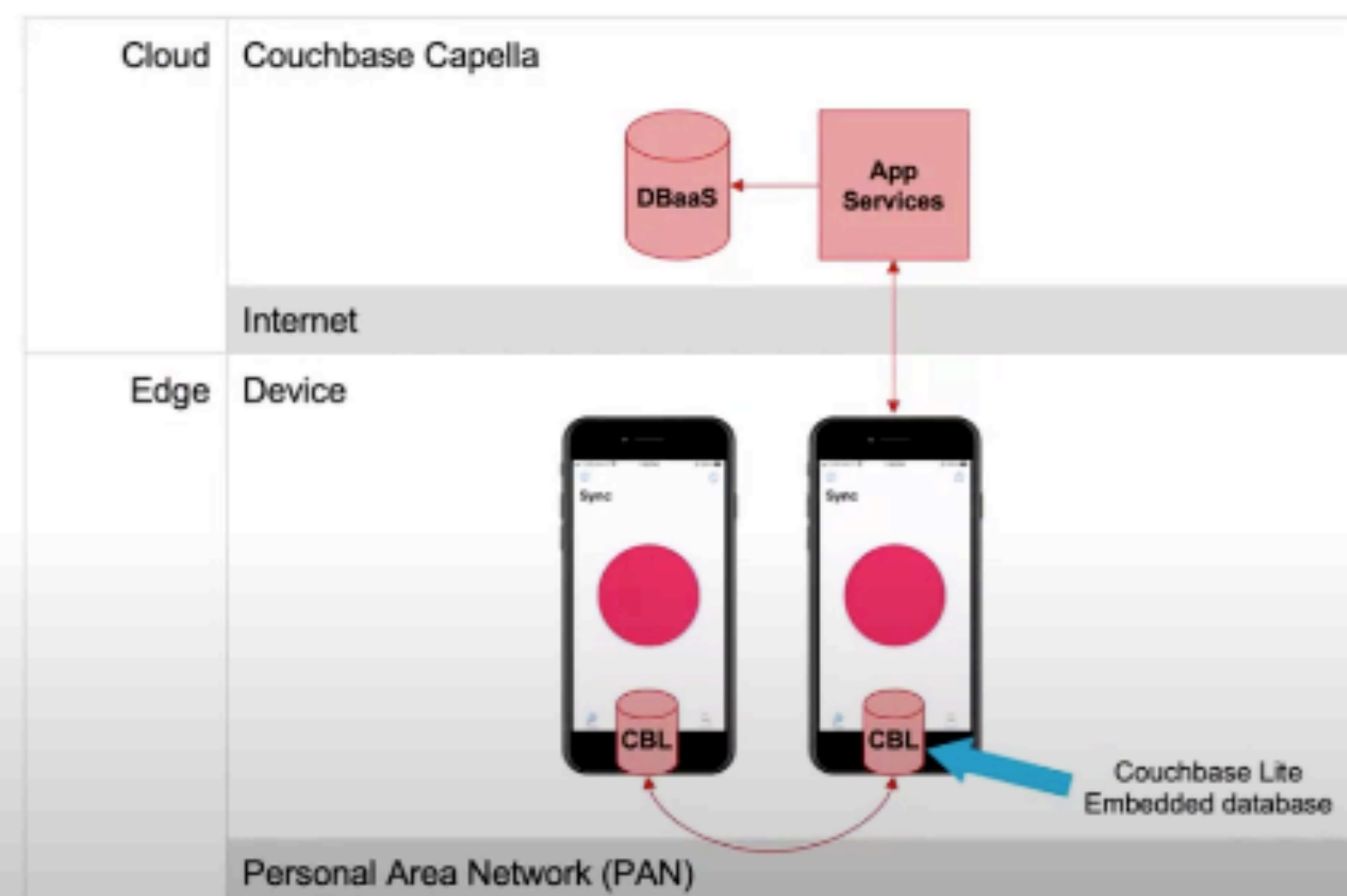
Embedded database
+ vector search
(offline-first!)

Devices running Couchbase Lite (Peer-to-Peer sync)



Mark Gamble

Demo architecture



0:54 / 7:16 • Demo architecture >



Couchbase Mobile - live data sync demo



Couchbase
7130位訂閱者

訂閱

15



分享

下載



